

20260202EB_MN (TDP43, NCT, IF, Hylight)

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2026年2月2日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 2 plates:

- 3 wells of iPSCs using accutase
- 4 wells of iPSCs using ReleSR

keep some iPSCs

2026年2月4日 星期三

T2 -> T3

2026年2月6日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年2月9日 星期一

T4 -> T5

2026年2月11日 星期三

T5 -> T6

2026年2月13日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年2月16日 星期一

T6 -> T7

Dissociation

1. Collect EBs into 15mL tubes (each plate). Wash the plate with 2mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 1mL 0.25% trypsin (prepare 8mL 0.25% trypsin with 100U DNaseI)
3. Pipette *5 with P1000 and then incubate the tubes in 37C.
4. Transfer 500uL pupp. to another tube containing FBS
5. add 1mL fresh trypsin -> pipette*10 at 8min -> 37C
6. pipette*10 at 12min -> add 1mL trypsin -> pipette*10 at 15min -> 37C (I can still see EBs)
7. pipette*10 at 20min (most EBs are dissolved)

8. transfer cells into 50mL tube containg 6mL FBS (as mentioned before) -> pipette with 5mL pipet ~ 10 times
9. add 6mL CTWM -> pipette ~ 10 times
10. filter cells through 40um cell strainer into new tubes.
11. centrifuge at 200xg, 4min

cell info ^

	A	B
1	cell	passage
2	WT11	P5

Table1 ^

	A	B
1		volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6

Table3

	A	B	C	D	E	F	G	H	I	J
1		conc. (K/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	accutase plate	756	without trypan blue	5	3780	A	24	225	24	6
3	ReleSR plate	1010	without trypan blue	5	5050	B	6	1143	3	the rest
4				total	8830					

Plate A ^

	1	2	3	4	5	6
A	12 uL in total (110)					
B	18 uL in total (111)					
C	0.5 (180)	1.5	5			
D	0.5 (181)	1.5	5			

Plate B				^
	1	2	3	
A				
B				

2026年2月17日 星期二

Change T7 -> T7 if added virus

2026年2月18日 星期三

T7-> T7 if not added virus

2026年2月19日 星期四

T7 -> T8

2026年2月22日 星期日

1/2 T8 -> 1/2 T8

2026年2月24日 星期二

1/2 T8 -> 1/2 T8

2026年2月25日 星期三

Imaging Hylight and TDP43

Treat

JFX650

stock:1mM

working:1nM

1. dilute 1mM JFX650 1:400 (0.5uL in 200uL) in no-glu media
2. add 1:2500 into 2.5mM or 0.15mM glucose media
 - a. 0.72 in 1.8mL 2.5mM gluc
 - b. 1.44 in 3.6mL 0.15mM gluc

Imaging note:

TDP43

- I accidentally changed A4 into 0.15mM for the 2nd treatment around 2min. I rinsed it with 2.5mM gluc and changed it back.
- I swapped A5 and A6 conditions order (originally A5 -> 2; A6 -> 3)

Hylight

- A lot of debris. It's a bit hard to find a cells with diffused signal. Lots of cells have very bright puncta inside.
- vLH180
 - 1.5uL vLH180 is the best. I can find some cells with diffused signal, but there was still a lot of debris.
 - 0.5uL had very few cells with diffused signal.
 - 5uL cells died.
- vLH181
 - in general have stonger 405 signal
 - 0.5uL have no diffused signal
 - 1.5uL almost nothing. I can only find one cell with diffused signal
 - D3 is the best one. I can find some diffused signal, but still a lot of debris.
- Treat cells with 0.5uM Rot/AA for 20min and then treat cells with 2DG for 20min.
 - Rot/AA: 15:07
 - Image: 15:43 (C2); 15:51 (D2) (reused the positions)
 - 2DG: 15:55
 - Image: 16:19 (C2); 16:14 (D2) (reused the positions)

Prepare Rot/AA and 2DG in BP T8

drug calculation																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1		stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (uM)	loading volume	well volume before loaded	loading dilution	loading conc.(uM)	dilution factor from 1st dil.	well# for each condition	total loading volume	volume of 1st dil.	volume of media	
2	Rotenone	5000	10	10	1	9	500	0.5	50	300	7	3.5	142.8571428571	2	125	0.87	123.7	
3	Antimycin A	1000	1				1000	0.5	50	300	7	3.5	285.7142857143	2	125	0.44		
4	2DG	2000000	1				2000000	50000	50	350	8	400000	5	2	125	25.00	100.0	
5	MMS	11800000	100	100	1	99	118000	5000	50	300	7	35000	3.3714285714	1	100	29.66	70.3	
6		11800000	100	100	1	99	118000	2000	50	300	7	14000	8.4285714286	1	100	11.86	88.1	
7		11800000	100	100	1	99	118000	500	50	300	7	3500	33.7142857143	1	100	2.97	97.0	
8		stock	working	dilution	# of well	media for each well (uL)	total volume (uL)	drug volume (uL)										
9	Leptomycin B	5.6 mg/mL	10 ng/mL	560	1	600	600	1.1										

condition#		
	A	B
1	condition#	
2	1	2.5 mM > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h -> 2.5mM
3	2	2.5 mM > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) -> 0.15mM
4	3	2.5 mM > 0.15 mM 1h > 0.15mM 3h > 0.15mM 1h > 0.15mM 3h > 0.15mM 1h -> 0.15mM
5	4	2.5mM
6	5	0.15mM
7	6	0.15mM+Ver150nM

live imaging condition							^
	1	2	3	4	5	6	
A	1	2	3	1	3	2	
B	1	2	3	4	5	6	
C							
D							

6-well conditions				^
	1	2	3	
A	1	2	3	
B				

time table for treatment																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1				treat#1			treat#2		treat#3		treat#4		treat#5				
2				0h		1h			4h		5h		8h		9h	24h	
3	condition#		condition	media	time	imaging	media after imaging	time	media	time	media	time	media	time	imaging	imaging on 2/28	
4	on 2/27/2026		110														IF
5	1	A1	2.5mM glucose constant	2.5mM gluc	9:30	10:30	2.5mM gluc	11:00	2.5mM gluc	13:30	2.5mM gluc	14:30	2.5mM gluc	17:30	18:30	9:11	
6	2	A2	0.15mM glucose recurrent	0.15mM gluc	10:00	11:00	2.5mM gluc	11:20	0.15m M gluc	14:00	2.5mM gluc	15:03	0.15mM gluc	18:00	19:00	9:37	
7	3	A3	0.15mM glucose constant	0.15mM gluc	10:30	11:30	0.15mM gluc	11:50	0.15m M gluc	14:30	0.15mM gluc	15:30	0.15mM gluc	18:30	19:36	10:14	
8			111														
9	1	B1	2.5mM glucose constant	2.5mM gluc	11:30	12:30	2.5mM gluc	12:50	2.5mM gluc	15:30	2.5mM gluc	16:30	2.5mM gluc	19:33	20:29	10:44	
10	2	B2	0.15mM glucose recurrent	0.15mM gluc	12:00	13:00	2.5mM gluc	13:20	0.15m M gluc	16:00	2.5mM gluc	17:00	0.15mM gluc	19:56	20:56	11:17	
11	3	B3	0.15mM glucose constant	0.15mM gluc	12:30	13:30	0.15mM gluc	13:50	0.15m M gluc	16:30	0.15mM gluc	17:30	0.15mM gluc	20:28	21:30	12:00	11:45-50 treat
12	on 2/25																12:45 fix
13	TDP43 on 2/25/2026															2/26/26	
14		A4	2.5mM glucose constant	2.5mM gluc + JFX650	11:00	12:08	2.5mM gluc	12:40	2.5mM gluc	15:15	2.5mM gluc	16:15	2.5mM gluc + JFX650	19:15	20:17	11:39	(add JFX650 at 10:30)
15		A5	0.15mM glucose constant	0.15mM gluc + JFX650	11:00	12:15	0.15mM gluc	12:40	0.15m M gluc	15:15	0.15mM gluc	16:15	0.15mM gluc + JFX650	19:15	20:25	11:42	(add JFX650 at 10:30)
16		A6	0.15mM glucose recurrent	0.15mM gluc + JFX650	11:00	12:23	2.5mM gluc	12:40	0.15m M gluc	15:15	2.5mM gluc	16:15	0.15mM gluc + JFX650	19:15	20:32	11:46	(add JFX650 at 10:30)
17	6-well plate				11:30	12:30				15:30		16:30		19:37	20:30		
18																	
19					treat	1h image	9h image	24h image on 2/26									
20	TDP43																
21	4	B4	2.5mM	2.5mM gluc	11:20	12:30	20:53	11:53									
22	5	B5	0.15mM	0.15mM gluc	11:20	12:49	21:01	12:02									
23	6	B6	0.15mM+LMB18.5 nM		11:20	12:57	21:10	12:12									
24																	
25																	
26																	

20260225_timetable													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		C1/C2	D1/D2	A4		A5		A6		6-well	B4	B5	B6
2	9:00			treat SPY505		treat SPY505		treat SPY505					
3	9:30												
4	10:00												
5	10:30												
6	11:00	image baseline		treat#1	2.5mM +JFX	treat#1	0.15mM +JFX	treat#1	0.15mM +JFX	treat#1			
7	11:30										treat 2.5mM	treat 0.15mM	treat 0.15mM + 18.5nM LMB
8	12:00			image@1h		image@1h		image@1h		treat#2			
9	12:30			treat#2	2.5mM gluc	treat#2	0.15mM gluc	treat#2	2.5mM gluc		image@1h	image@1h	image@1h
10	13:00												
11	13:30												
12	14:00												
13	14:30												
14	15:00	Rot/AA	Rot/AA	treat#3	2.5mM gluc	treat#3	0.15mM gluc	treat#3	0.15mM gluc	treat#3			
15	15:30	image	image										
16	16:00	2DG	2DG	treat#4	2.5mM gluc	treat#2	0.15mM gluc	treat#4	2.5mM gluc	treat#4			
17	16:30	image	image										
18	17:00												
19	17:30												
20	18:00												
21	18:30												
22	19:00			treat#5	2.5mM +JFX	treat#5	0.15mM +JFX	treat#5	0.15mM +JFX	treat#5			
23	19:30												
24	20:00			image@9h		image@9h		image@9h		collect (9h)			

TDP43_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	16	10	200 ms	The first time I used SPY505. The signal is great
3	640	16	10	300 ms	

fileid_treat_TDP43

	A	B	C	D	E
1	image	treatment	treat_time	type	drug
2	MN_TDP43Halo_60x_0001	2.5mM gluc	1h	constant	
3	MN_TDP43Halo_60x_0002	0.15mM gluc	1h	constant	
4	MN_TDP43Halo_60x_0003	0.15mM gluc	1h	recurrent	
5	MN_TDP43Halo_60x_0004	2.5mM gluc	1h	no-change	
6	MN_TDP43Halo_60x_0005	0.15mM gluc	1h	no-change	
7	MN_TDP43Halo_60x_0006	0.15mM gluc	1h	no-change	LMB-18.5nM
8	MN_TDP43Halo_60x_0007	2.5mM gluc	9h	constant	
9	MN_TDP43Halo_60x_0008	0.15mM gluc	9h	constant	
10	MN_TDP43Halo_60x_0009	0.15mM gluc	9h	recurrent	
11	MN_TDP43Halo_60x_0010	2.5mM gluc	9h	no-change	
12	MN_TDP43Halo_60x_0011	0.15mM gluc	9h	no-change	
13	MN_TDP43Halo_60x_0012	0.15mM gluc	9h	no-change	LMB-18.5nM
14	MN_TDP43Halo_60x_0013	2.5mM gluc	24h	constant	
15	MN_TDP43Halo_60x_0014	0.15mM gluc	24h	constant	
16	MN_TDP43Halo_60x_0015	0.15mM gluc	24h	recurrent	
17	MN_TDP43Halo_60x_0016	2.5mM gluc	24h	no-change	
18	MN_TDP43Halo_60x_0017	0.15mM gluc	24h	no-change	
19	MN_TDP43Halo_60x_0018	0.15mM gluc	24h	no-change	LMB-18.5nM

Hylight_O.C. , 60X, 600x900, 16-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	16	30	500 ms	signal is not strong for both. Also, A LOT of debris.
3	Dapi_ex_Green_e m	16	20	400 ms	

fileid_treat_Hylight			
	A	B	C
1	image	treatment	time
2	vLH180_1.5_60x	NT	0h
3	vLH180_1.5_60x_0001	Rot/AA	30min
4	vLH180_1.5_60x_0002	2DG	20min
5	vLH181_5_60x	NT	0h
6	vLH181_5_60x_0001	Rot/AA	30min
7	vLH181_5_60x_0002	2DG	20min



2026年2月26日 星期四

- ☒ BP T8
- ☒ Pre-treat 1:6000 SirDNA for 1 hour from 16:00-17:00 (to avoid toxicity)

Image 24h of TDP43

20260226_timetable										
	A	B	C	D	E	F	G	H	I	J
1		A4		A5		A6		B4	B5	B6
2	10:30	treatJFX		treatJFX		treatJFX		treatJFX	treatJFX	treatJFX
3	11:00									
4	11:30	image@24h		image@24h		image@24h		image@24h	image@24h	image@24h

Cells behaved differently with 1h treatment of hypoglycemia. The 2.5mM gluc no-change group seems to have a very low N/C ratio at the beginning.

2026年2月27日 星期五

Imaging: NCT

Note:

- The signal of this batch is very good. Cells looked great as well.
- SirDNA signal has more very bright punta around the nucleus (more than 1h) at 9h. May consider treating the cells on the same day next time.

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	The signal for this batch is much better and higher!
3	640	16	10	100 ms	
4	488	16	50	1s	

fileID_treatment_NCT

	A	B	C	D	E	F
1	image	reporter	treatment	time	type	note
2	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	2.5mM gluc	1h	constant	
3	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	2.5mM gluc	1h	constant	
4	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.15mM gluc	1h	recurrent	
5	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.15mM gluc	1h	recurrent	
6	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0.15mM gluc	1h	constant	
7	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.15mM gluc	1h	constant	
8	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2.5mM gluc	9h	constant	
9	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2.5mM gluc	9h	constant	position one is out of focus for a frame
10	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	0.15mM gluc	9h	recurrent	
11	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	0.15mM gluc	9h	recurrent	
12	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	0.15mM gluc	9h	constant	
13	com_110_MN_no-glu_50%_1s_30s_60x_0012	110	0.15mM gluc	9h	constant	
14	com_110_MN_no-glu_50%_1s_30s_60x_0013	110	2.5mM gluc	24h	constant	
15	com_110_MN_no-glu_50%_1s_30s_60x_0014	110	2.5mM gluc	24h	constant	
16	com_110_MN_no-glu_50%_1s_30s_60x_0015	110	0.15mM gluc	24h	recurrent	
17	com_110_MN_no-glu_50%_1s_30s_60x_0016	110	0.15mM gluc	24h	recurrent	
18	com_110_MN_no-glu_50%_1s_30s_60x_0017	110	0.15mM gluc	24h	constant	
19	com_110_MN_no-glu_50%_1s_30s_60x_0018	110	0.15mM gluc	24h	constant	
20	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	2.5mM gluc	1h	constant	
21	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	2.5mM gluc	1h	constant	
22	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.15mM gluc	1h	recurrent	
23	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.15mM gluc	1h	recurrent	
24	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0.15mM gluc	1h	constant	
25	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.15mM gluc	1h	constant	
26	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2.5mM gluc	9h	constant	
27	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2.5mM gluc	9h	constant	
28	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	0.15mM gluc	9h	recurrent	
29	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	0.15mM gluc	9h	recurrent	
30	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	0.15mM gluc	9h	constant	
31	com_111_MN_no-glu_50%_1s_30s_60x_0012	111	0.15mM gluc	9h	constant	
32	com_111_MN_no-glu_50%_1s_30s_60x_0013	111	2.5mM gluc	24h	constant	
33	com_111_MN_no-glu_50%_1s_30s_60x_0014	111	2.5mM gluc	24h	constant	
34	com_111_MN_no-glu_50%_1s_30s_60x_0015	111	0.15mM gluc	24h	recurrent	
35	com_111_MN_no-glu_50%_1s_30s_60x_0016	111	0.15mM gluc	24h	recurrent	
36	com_111_MN_no-glu_50%_1s_30s_60x_0017	111	0.15mM gluc	24h	constant	
37	com_111_MN_no-glu_50%_1s_30s_60x_0018	111	0.15mM gluc	24h	constant	

2026年2月28日 星期六

IF condition							^
	1	2	3	4	5	6	
A				MMS 0.5mM			
B				MMS 2mM			
C				MMS 5mM	4	4	
D				MMS 0	5	5	

Methyl methanesulfonate (MMS)

Molecular Weight: 110.13
density: 1.3 g/mL at 25 °C (lit.)
--> 11.8M
300ul of BP T8 for each well.
add 50uL drug

1. Treat cells (C5,6 & D5,6) with different glucose + Halo-dye (10nM) for 1h / column4 with MMS
- a. 0.44uL 25uM in 1100uL media
2. Prepare 4%PFA (8 wells: 200uL/well. 1400 uL PBS + 200 uL 32%PFA)
3. ~~Wash cells with cold PBS at~~ Only remove the media
4. Fix cells with 4% PFA for 15 min
5. Wash with PBS
6. Permeabilization with 0.2% triton (100uL 5% triton in 2400uL PBS) for 10min (I didn't permeabilize D1-3, but permeablize all the rest)
7. Blocking with 10% FBS for overweekend in cold room.

2026年3月6日 星期五

Imaging IF

Cells look good. They attached well on the glass.
Old Spinning disk microscope

SetA OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC			BF
3	405	10	50	DAPI
4	488	50	100	TDP43
5	561	50	500	Importin-b1
6	640	20	100	Halodye (TDP43-Halo)
7	z-stack	0.5 um/step	10	21 steps in total

SetB OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC		100	BF
3	405	10	50	DAPI
4	488	50	100	RANGAP1
5	50	50	300	Ran
6	640		100	Halodye (TDP43-Halo)
7	z-stack	0.5 um/step	10	21 steps in total

Table7

	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	360	0.72	488	B	1000	600	0.6
3	A	147	Importin-b1	m	1000	360	0.36	594	O	1000	1200	1.2
4	B	C-31	RANGAP1	Rb	500	360	0.72	488	B			
5	B	146	Ran	m	500	360	0.72	594	O			
6	C	63	H2A.X (Ser139) clone JBW301	Mouse mono	2000	720	0.36	594	O			

fileID_treatment_IF						
	A	B	C	D	E	F
1	image	treatment	treat_time	488	561	640
2	C6_488-TDP43_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	TDP43	Impb1	TDP43Halo
3	D6_488-TDP43_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	TDP43	Impb1	TDP43Halo
4	C5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	RanGAP1	Ran	TDP43Halo
5	D5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	RanGAP1	Ran	TDP43Halo